

[Link to the Gameplay Recording](#)

Puzzle 1: Summary

This puzzle involves quickly presenting the player with a stimulus (a colored object) on either the left or right side of the screen. The player must press the corresponding color-coded button if the object is blue or red, regardless of its location before the stimulus disappears. The location of the stimulus is irrelevant, but it can overlap with the player's response location (e.g., a blue stimulus on the left might tempt the player to associate it with the left-side response). If the player does not respond before stimulus disappears or presses an incorrect button, it is considered a failure. Along with that, player must not press any button with the appearance of a green object as a stimulus.



the blue button is on the top right corner and the red button is on the top left corner of the keyboard

There are 3 practice trials and 10 actual trials. If the player accurately reacts to the given stimulus, the player scores a point. Each point the player earned will be used as “extra trials” throughout the escape room. There is no “one” solution to this game. Better performance on this game allows the team to gain more opportunities (extra trials) to attempt other challenges within the escape room, increasing their chances of overall success.

Puzzle 1: Theoretical Principles

Simon Effect

The task incorporates the Simon Effect, where the irrelevant location of a stimulus interferes with response selection. When the spatial location of the cue (left or right) aligns with the response location (button press), reaction times are faster, while misalignment increases reaction times and error rates. This demonstrates how congruity between irrelevant and relevant features affects cognitive performance.

Players will exhibit higher accuracy when the spatial location of the stimulus (left or right) aligns with the button's location (congruent trials), while higher error rates when the stimulus location conflicts with the button's location (incongruent trials).

Stop-Signal Paradigm

The requirement to suppress responses to the green cue reflects response inhibition. This aligns with the Stop-Signal Paradigm, which emphasizes stopping automatic responses. The puzzle challenges players to inhibit automatic reactions triggered by visual stimuli. Players will occasionally fail to suppress their responses to the green stimulus during rapid sequences, reflecting the difficulty of overriding an automatic response to visual stimuli.

Puzzle 2: Summary

This puzzle involves three players collaborating to interpret and respond to a shared stimulus.

Players 1 and 2 observe an 8-digit sequence (composed of digits 1–4) displayed on the screen, where each digit is shown in either blue or red font.

- **Player 1:** Observes a word ("blue" or "red") displayed in either blue or red font.

Regardless of the font color, Player 1 must respond to the word's content:

- "Blue": Perform an upward hand gesture.
- "Red": Perform a downward hand gesture.

- **Player 2:** Observes a single-digit number (1–4) displayed in black font at the top of the screen. Using this number, Player 2 determines how many black font digits displayed at the top of the screen are present in the 2 rows of 10-digit sequence displayed on the screen. This sequence consists of digits shown in red, blue, and black fonts.

- **Player 3:** Integrates the inputs from Players 1 and 2. Based on Player 1's gesture, Player 3 selects the position (up or down) and presses the number Player 2 calculates.



The “top” digits (1,2,3,4) are the four buttons next to the red button.

The “down” digits (1,2,3,4) are the four buttons in burgundy color.

For example:

- Player 1 observed **BLUE**
- Player 2 observed target digit as 1 and the given 20 digits are:
[1234123412]
[1234123412]
- Player 3 must press button 3 on the top because the content of the word player 1 observes is blue (upward hand gesture) and there are 3 black font digit 1.

If the team achieves more than 80% accuracy, they can proceed to the next room.

Puzzle 2: Theoretical Principles

Selective Attention

Both Players 1 and 2 rely on selective attention to process their respective task-relevant features while ignoring distractions:

- **Player 1:** Focuses on the word's content ("blue" or "red") while ignoring the font color. This aligns with filter theories of attention, where irrelevant information is suppressed to focus on task-relevant details.
- **Player 2:** Must attend to the black font number at the top and accurately count the black font target digit in the 2 rows of 10-digit sequence, ignoring digits in blue and red fonts. This highlights attentional filtering under high perceptual load.

Players 1 and 2 may show slower response times or errors when the irrelevant features (e.g., font color for Player 1 or red/blue digits for Player 2) are highly salient, illustrating the cognitive costs of filtering irrelevant information.

Negative Priming

Player 2 also experiences negative priming because the task requires them to focus on specific black font digits in one trial while ignoring others. For example, if the digit "2" in black font was counted as relevant in the previous trial but is now irrelevant, the lingering cognitive processing from the prior trial (residual inhibition) may interfere with ignoring it in the current trial.

Conversely, a black font digit like "4" that was ignored before might now be task-relevant, causing a delay as the inhibitory effect is overcome.

Player 2 may demonstrate increased response times or errors when a previously counted black font digit (e.g., "2") becomes irrelevant or a previously ignored black font digit (e.g., "4") becomes relevant. These effects are due to the cognitive cost of switching attentional focus between relevant and irrelevant digits across trials.

Psychological Refractory Period

Player 3 is responsible for combining inputs from Players 1 and 2 and performing the appropriate action based on this information. This task aligns with the psychological refractory period (PRP), a phenomenon where dual-task interference arises due to a processing bottleneck. Player 3 must process Player 1's gesture and Player 2's calculated number sequentially. A bottleneck occurs because Player 3 cannot process Player 2's information until the response selection for Player 1's input is complete, or vice versa.

- When Player 1's gesture requires additional time for interpretation (e.g., due to ambiguous or conflicting cues), it delays Player 3's ability to select and execute the correct response based on Player 2's calculation.
- The PRP effect demonstrates that although perception and motor preparation can happen simultaneously, response selection must occur in a sequential manner. This serial processing creates interference when both tasks overlap in time.

Player 3 may experience slower reaction times or errors when inputs from Players 1 and 2 arrive simultaneously or are unclear, illustrating the attentional bottlenecks. However, upon multiple trials, player 3 will exhibit faster reaction times due to practice. There are several mechanisms that can explain this. Player 3 may have become more efficient (task

automatization), reduced the time required to complete individual tasks (stage shortening), or combined the two tasks into a single task(task integration) through practice.